

Expense Tracker Application – Java console - based project

1. Introduction

Managing daily expenses is an important part of personal financial planning. Manual tracking of expenses can become difficult as the number of transactions increases. To address this problem, the Expense Tracker Application was developed as a Java-based console application that allows users to record, manage, and analyze their expenses efficiently.

The application provides functionalities such as adding expenses, viewing all recorded expenses, generating category-wise summaries, comparing monthly spending patterns, and deleting expense records. The project demonstrates the practical application of core Java programming concepts.

2. Problem Statement

Many individuals find it difficult to keep track of their daily expenses and spending habits. Without proper expense management, it becomes challenging to identify spending patterns and control unnecessary expenditures.

This project aims to provide a simple and effective solution for recording and analyzing personal expenses through a menu-driven application.

3. Objectives

The main objectives of the project are:

- To record daily expenses.
- To store expense records permanently.
- To display expense history.
- To generate category-wise expense summaries.
- To compare monthly spending trends.
- To provide an easy-to-use console interface.

4. Technologies Used

Technology – it's purpose

Java - Application Development

OOP Concepts - Program Structure

ArrayList - Expense storage

HashMap - Expense Analysis

File Handling - Data Persistence

Serialization - Saving Objects

Exception Handling - Error Management

Date APIs - Monthly Comparison

5. System Design

The application follows a simple object-oriented design consisting of three main classes.

Expense Class

The Expense class represents an individual expense record.

Attributes:

- Date
- Category
- Amount
- Note

Responsibilities:

- Store expense details.
- Provide getter methods.
- Display expense information.

ExpenseManager Class

The ExpenseManager class performs all expense management operations.

Responsibilities:

- Add expenses.
- Display expenses.
- Generate category summaries.
- Generate monthly comparisons.
- Delete expenses.
- Save data to file.
- Load data from file.

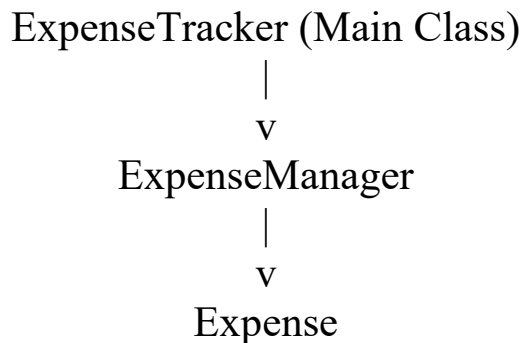
ExpenseTracker Class

This is the main class of the application.

Responsibilities:

- Display menu options.
- Accept user input.
- Invoke required operations.
- Control program execution.

6. System Architecture



The ExpenseTracker class interacts with the user, ExpenseManager processes the data, and Expense stores individual expense information.

7. Features

Add Expense

Allows users to enter:

- Date
- Category
- Amount
- Optional Note

The entered expense is stored in memory and saved to a file.

View All Expenses

Displays all recorded expenses in a numbered format.

Category-wise Summary

Groups expenses based on category and calculates the total amount spent in each category.

Example:

Food : Rs.2500

Travel : Rs.1500

Shopping : Rs.3000

Monthly Comparison

Calculates monthly expense totals and compares the latest month with the previous month.

The feature helps users understand spending trends and identify changes in expenditure.

Delete Expense

Allows users to remove an existing expense record by selecting its corresponding number.

Persistent Storage

All expense records are stored in a file named:

expenses.dat

Data remains available even after the application is closed and reopened.

8. Concepts Implemented

Object-Oriented Programming

The project demonstrates:

- Classes
- Objects
- Encapsulation
- Method-based modular programming

Collections Framework

ArrayList

Used to store expense records dynamically.

HashMap

Used to generate:

- Category-wise summaries
- Monthly expense summaries

File Handling

The project uses:

- ObjectOutputStream
- ObjectInputStream

These classes help store and retrieve expense data from a file.

Exception Handling

Try-catch blocks are used to handle:

- File-related errors
- Invalid date formats
- Unexpected runtime issues

Date Processing

The project uses:

- SimpleDateFormat
- Calendar

These classes are used to extract month and year information for monthly analysis.

9. Sample Execution

Menu Options:

1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit

Users can select an option and perform the corresponding operation.

```
C:\Users\B DEEKSHITHA>cd C:\Users\B DEEKSHITHA\OneDrive\Desktop\2024
C:\Users\B DEEKSHITHA\OneDrive\Desktop\2024>javac ExpenseTracker.java
C:\Users\B DEEKSHITHA\OneDrive\Desktop\2024>java ExpenseTracker.java

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit
Choose an option: 2

All Recorded Expenses:
1. Date: 01-02-2025, Category: Bill ele, Amount: Rs.2000.0, Note: -
2. Date: 04-02-2025, Category: Travel, Amount: Rs.2500.0, Note: home

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit
Choose an option: 1
Enter Date (DD-MM-YYYY): 02-03-2025
Enter Category: food
Enter Amount (Rs.): 1250
Enter Note (optional): outing
Expense added successfully!

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
```

EXPENSE TRACKER APPLICATION

```
5. Delete Expense
6. Exit
Choose an option: 1
Enter Date (DD-MM-YYYY): 05-03-2025
Enter Category: shopping
Enter Amount (Rs.): 1800
Enter Note (optional): bday shopping
Expense added successfully!

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit
Choose an option: 1
Enter Date (DD-MM-YYYY): 05-04-2025
Enter Category: hostel fee
Enter Amount (Rs.): 50000
Enter Note (optional): -
Expense added successfully!

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit
Choose an option: 2

All Recorded Expenses:
1. Date: 01-02-2025, Category: Bill ele, Amount: Rs.2000.0, Note: -
2. Date: 04-02-2025, Category: Travel, Amount: Rs.2500.0, Note: home
3. Date: 02-03-2025, Category: food, Amount: Rs.1250.0, Note: outing
4. Date: 05-03-2025, Category: shopping, Amount: Rs.1800.0, Note: bday shopping
5. Date: 05-04-2025, Category: hostel fee, Amount: Rs.50000.0, Note: -
```



```
===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit
Choose an option: 3

Expense Summary by Category:
hostel fee : Rs.50000.0
bill ele : Rs.2000.0
travel : Rs.2500.0
food : Rs.1250.0
shopping : Rs.1800.0

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit
Choose an option: 4

Total Expenses by Month:
2-2025 : Rs.4500.0
3-2025 : Rs.3050.0
4-2025 : Rs.50000.0

Comparison:
Last Month (3-2025): Rs.3050.0
This Month (4-2025): Rs.50000.0
You spent Rs.46950.0 more this month

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
```

```

5. Delete Expense
6. Exit
Choose an option: 5

All Recorded Expenses:
1. Date: 01-02-2025, Category: Bill ele, Amount: Rs.2000.0, Note: -
2. Date: 04-02-2025, Category: Travel, Amount: Rs.2500.0, Note: home
3. Date: 02-03-2025, Category: food, Amount: Rs.1250.0, Note: outing
4. Date: 05-03-2025, Category: shopping, Amount: Rs.1800.0, Note: bday shopping
5. Date: 05-04-2025, Category: hostel fee, Amount: Rs.50000.0, Note: -
Enter the number of the expense to delete: 3
Deleted: Date: 02-03-2025, Category: food, Amount: Rs.1250.0, Note: outing

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit
Choose an option: 4

Total Expenses by Month:
2-2025 : Rs.4500.0
3-2025 : Rs.1800.0
4-2025 : Rs.50000.0

Comparison:
Last Month (3-2025): Rs.1800.0
This Month (4-2025): Rs.50000.0
You spent Rs.48200.0 more this month

===== Expense Tracker Menu =====
1. Add Expense
2. View All Expenses
3. View Summary by Category
4. View Monthly Comparison
5. Delete Expense
6. Exit
Choose an option: 6
Exiting and saving data...

```

10. Advantages

- Easy to use.
- Lightweight console application.
- Permanent storage of expense records.
- Monthly spending analysis.
- Category-wise expense tracking.
- Demonstrates important Java concepts.

11. Limitations

- Console-based interface.
- Single-user application.

- No database integration.
- No graphical reports.
- Limited search and filtering options.

12. Future Enhancements

Possible future improvements include:

- Search Expense functionality.
- Edit Expense functionality.
- Budget tracking.
- Graphical User Interface (GUI).
- Database integration using MySQL or SQLite.
- Graph and chart visualization.
- Export reports to PDF or Excel.

13. Conclusion

The Expense Tracker Application successfully demonstrates the practical implementation of Java programming concepts including Object-Oriented Programming, Collections Framework, File Handling, Exception Handling, and Date Processing.

The application provides an effective solution for managing personal expenses while helping developers gain experience in building real-world Java applications. The project serves as a strong foundation for future enhancements such as graphical interfaces, databases, and advanced financial analysis features.